

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
Department of Computer Science & Engineering
End Semester/Supplementary Examination, Summer Semester 2024
B.Tech. 6th Semester

Subject Name: Computer Networks
Subject Code: CS3002
Number of pages:2

Maximum Marks: 50
Exam Duration: 3 Hours

Note: There are Seven questions. Attempt Any Five questions. All the parts of every question must be attempted in a sequence.

Q.1 A CRC is constructed to generate a 4-bit FCS for an 11-bit message. The generator polynomial is $X^4 + X^3 + 1$.

- a.) Encode the data bit sequence 10011011100 using the generator polynomial and give the codeword
- b.) Now assume that bit 7 (counting from the leftmost bit) In codeword is in error and show that the detection algorithm detects the error

[5+5=10 Marks]

Q.2 A receiver receives the code 11001100111. When it uses the Hamming encoding algorithm, the result is 0101. Which bit is in error? What is the correct code?

[10 Marks]

Q.3 A large population of ALOHA users manages to generate 50 requests/sec, including both originals and retransmissions. Time is slotted in units of 40 msec.

- a. What is the chance of success on the first attempt?
- b. What is the probability of exactly k collisions and then a success?
- c. What is the expected number of transmission attempts needed?

[3+3+4=10 Marks]

Q.4(a) Consider a 10-Mbps Ethernet LAN that has stations attached to a 205 km long coaxial cable. Given that the transmission speed is 2.3×10^8 m/s in a coaxial cable, then find out the throughput of system and taking packet size as 128 bytes.

(b) ARP and RARP both map addresses from one space to another. In this respect, they are similar. However, their implementations are fundamentally different. In what major way do they differ?

[5+5=10 Marks]

Q.5 (a) Write the seven layers of OSI Model. Also draw the architecture of the OSI model.

(b) Differentiate between non-persistent HTTP and persistent HTTP.

[5+5=10 Marks]

Q.6 Suppose that host A is connected to a router R 1, R 1 is connected to another router, R 2, and R 2 is connected to host B. Suppose that a TCP message that contains 900 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to B. Show the Total length, Identification, DF, MF, and Fragment offset fields of the IP header in each packet transmitted over the three links. Assume that link A-R1 can support a maximum frame size of 1024 bytes including a 14-byte frame header, link R1-R2 can support a maximum frame size of 512 bytes, including an 8-byte frame header, and link R2-B can support a maximum frame size of 512 bytes including a 12-byte frame header.

[10 Marks]

Q.7 Consider a 100Mbps 100BT Ethernet. In order to have an efficiency of .50, what should be the maximum distance between a node and the hub? Assume a frame length of 64 bytes and that there are no repeaters. Does this maximum distance also ensure that a transmitting node A will be able to detect whether any other node transmitted while A was transmitting? Why or why not? How does your maximum distance compare to the actual 100 Mbps standard?

[10 Marks]